

Precision vs. Accuracy

There are two types of quantities in science: exact values and measurements. Measurements always have some uncertainty or error associated with them. There are two types of measurement error: **Random error** is due to precision of the measuring instrument. For instance, the error on a 100 mL graduated cylinder may be ± 2 mL. A **systematic error** is associated with an inherent problem with the measuring system, room conditions, incorrect calibration, etc.

Precision is the place value of the last measurable digit. This means, the more decimal places, the more precise the reading. When adding and subtracting values, the answer is always rounded to the least precise measurement.

$$(11.7 \text{ cm} + 3.29 \text{ cm} + 0.542 \text{ cm}) = 15.532 \text{ cm}$$

15.532 cm would be rounded to 15.5 cm.

Ex) Solve

$$3.296 + 1.003 - 30.36 + 36.210$$

$$10.149$$

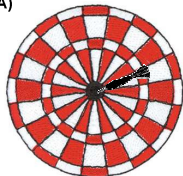
$$= 10.15$$

No matter how precise a measurement, it may still not be accurate. **Accuracy** refers to how close a value is to its true value. The comparison of the two values can be calculated by:

$$\% \text{ difference} = \frac{|\text{experimental} - \text{predicted}|}{\text{predicted}} \times 100\%$$

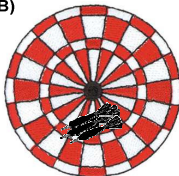
Dart Board Analogy

A)



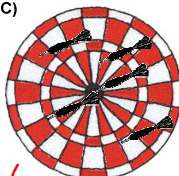
Accurate

B)



Precise

C)



Neither precise nor accurate